

AutoMind AI

Automotive Diagnostic Health Report

Generated by AutoMind AI 0.1.0 MVP

SIMULATION MODE - This report contains simulated data and is not a real vehicle diagnosis.

Vehicle Information

Vehicle	2024 AutoMind Demo Diagnostics Lab
VIN	SIMULATEDVIN000001
Engine	2.0L I4
Fuel Type	Gasoline
Diagnostic Date	2026-09-30T16:18:19.895580+00:00
Adapter / Mode	Built-in Vehicle Simulator
Health Summary	Attention Required

Detected Faults

Code	Status	Description	Severity
P0302	Stored	Cylinder 2 Misfire Detected	High

Live Data Summary

PID	Measurement	Value
010C	Engine RPM	697 rpm
010D	Vehicle Speed	0.0 km/h
0105	Engine Coolant Temperature	89.8 °C
010F	Intake Air Temperature	30.0 °C
0104	Calculated Engine Load	34.9 %
0111	Throttle Position	12.7 %
0110	MAF Air Flow Rate	3.44 g/s
010B	Intake Manifold Absolute Pressure	34.1 kPa
0106	Short-Term Fuel Trim Bank 1	7.7 %
0107	Long-Term Fuel Trim Bank 1	1.9 %
0114	O2 Sensor 1 Voltage	0.55 V
0142	Control Module Voltage	14.12 V
010E	Ignition Timing Advance	9.88 °
0146	Ambient Air Temperature	24.0 °C
012F	Fuel Tank Level Input	58.0 %

Freeze Frame

PID / Item	Value
010C - Engine RPM	1420 rpm
0105 - Engine Coolant Temperature	88.0 °C
0106 - Short-Term Fuel Trim Bank 1	10.2 %
0107 - Long-Term Fuel Trim Bank 1	4.1 %
0110 - MAF Air Flow Rate	8.8 g/s

Diagnostic Interpretation

P0302 - Cylinder 2 Misfire Detected

Observed evidence

- P0302 is reported as Stored: Cylinder 2 Misfire Detected.
- Fuel trims: STFT 7.7%, LTFT 1.9%.

Possible causes (not confirmed)

- Ignition coil - Common cause for this DTC
- Spark plug - Common cause for this DTC
- Injector - Common cause for this DTC
- Wiring - Common cause for this DTC
- Compression issue - Common cause for this DTC
- Vacuum leak affecting cylinder - Common cause for this DTC

Recommended checks

1. Check Cylinder 2 ignition coil
2. Compare or swap coil only where safe/appropriate
3. Inspect Cylinder 2 spark plug
4. Check injector operation
5. Review fuel trims and related codes
6. Perform compression/leak-down test if ignition/fuel checks pass

Confidence / uncertainty: Moderate: based on the reported DTC and available OBD-II measurements; confirm with vehicle-specific testing.

Warning: A severe active misfire can overheat and damage the catalytic converter. Avoid prolonged driving if the MIL is flashing or the engine runs very poorly.

Technician / User Notes

Demo note: rough idle reported during the simulated inspection.

Important Disclaimer

AutoMind AI is a diagnostic assistant. OBD-II codes and measured sensor values are evidence from the ECU or simulator; suggested mechanical causes are hypotheses that require appropriate inspection and testing. This report does not replace vehicle-specific service information, qualified diagnosis, or safe workshop procedures. Do not rely on this report for safety-critical repair decisions without verification.